

Concurrency budget: where a 70B host's 640 GB pool goes

AGGREGATE RESIDENCY SCREEN $\neq$ PER-RANK FIT GUARANTEE

~ 436 GB KV @ FP16 (2.62 MB/token, 9.5K max)

scenario reserve
(not a hardware constant)

FP16 KV



$C \approx 17.5$ concurrent requests @ FP16

FP8 KV



$8 \times H100 = 640$ GB

FP8 KV: ~ 32.4 slots ($436 \div 13.448$ GB/request)



8×80 GB per-rank pool (the aggregate screen $\neq$ a per-rank fit guarantee)

0 200 400 600 800
on-host HBM (GB, per $8 \times H100$ host)